

Erwin POUSSI

erwinpi@stanford.edu | US: +1 650 250 7491 | FR: +33 (0)7 49 62 90 59 | Website | GitHub | LinkedIn

MS in Aero/Astro at Stanford — robot learning, motion planning, LLM fine-tuning, sim-to-real transfer.

EDUCATION

Stanford University, Palo Alto (CA) 2025 – 2027

- MS in Aeronautics & Astronautics GPA: **4.1/4.0**
- Relevant coursework: *Principles of Robot Autonomy (AA274A/CS237A)*, *Introduction to Robotics (CS223A)*, *Decision Making under Uncertainty (AA228/CS238)*, *Optimal and Learning-Based Control (AA203)*, *Deep Learning & Reinforcement Learning (CS224R)*.

Ecole Polytechnique, Palaiseau (FRA) 2020 – 2025

- Master's level in Mechanical Engineering, concentration in Autonomy, minor in Entrepreneurship. GPA: **3.96/4.0**
- Relevant coursework: *Advanced Machine Learning for Autonomous Agents*, *Optimization and Control*, *Statistics*.

WORK EXPERIENCE

Stanford Multi-Robot Systems Lab Palo Alto, CA Jan. 2025 – Present

- Foundation Model Training — Imitation Learning & Sim-to-Real Transfer:** Trained vision-language-action (VLA) policies for autonomous drone navigation via behavioral cloning and DAGger.
- Implemented RRT* trajectory planning + MPC expert policy for data generation; grounded language instructions to 3D Gaussian Splat scenes via CLIPSeg; trained BC + DAGger policies, reaching **90% navigation success** (up from 52%). ([project page](#))
- GAIT — Agentic Scene Understanding for Open-Vocabulary Navigation** (Jun.–Sep. 2026): Built a graph-memory architecture for VLA models leveraging online semantic scene graph construction to enable open-vocabulary navigation with interrupt-triggered re-planning. ([project page](#))

Stanford School of Medicine, Qiu Lab Palo Alto, CA Oct. 2025 – Present

- ML Researcher — Pantheon-CLI:** Multi-agent LLM framework for autonomous scientific discovery with tool-augmented inference for genomic analysis.
- Implemented actor-critic RL for gene panel selection in scRNA-seq; fine-tuned Qwen3 LLMs via SFT and RLOO to power a self-evolving code-generation agent backbone. ([project page](#))
- Virtual Embryo Challenge — 3D Convolutional Models for Embryogenesis** (Jun.–Sep. 2026): Training foundational 3D convolutional models to reproduce embryogenesis, predicting cell-level gene expression dynamics and morphogenesis; co-building the [Virtual Embryo Challenge](#). ([project page](#))

Stevens Institute of Technology, RRG Hoboken, NJ Apr. 2025 – Aug. 2025

- Research Intern — ML-Based Aerodynamic Modeling:** Modeled parachute broadcloth permeability under rarefied conditions using a multi-fidelity ML framework combining random forests and genetic algorithms for parameter inference from experimental data; achieved **<1% mean error**. ([project page](#))

SELECTED RESEARCH

Reinforcement Learning for Algorithmic Trading Jan. 2025 – Apr. 2025

- Built a custom high-frequency market simulator and trained agents with PPO, Actor-Critic, and DQL, focusing on reward-free exploration, robustness to distribution shifts, and real-time decision-making.

Guidance System for Autonomous Visual-Based Docking — Nyx Spacecraft (CSEP) Sep. 2023 – May 2024

- Undergraduate thesis on vision-based navigation for the Nyx spacecraft (selected by Axiom for ISS support). Supervised by The Exploration Company with Polytechnique Space Center. ([project page](#))

TECHNICAL SKILLS

Programming & Frameworks: Python, PyTorch, CUDA, ROS2, MuJoCo, Hugging Face / Transformers, ACADOS, GSplat, GTSAM, CLIPSeg, Habitat-Sim, W&B

Methods: Behavioral cloning, DAGger, PPO, Actor-Critic, DQL, RRT*, MPC, SFT, RLOO, RLVR, LLM fine-tuning, imitation learning, sim-to-real transfer, motion planning, trajectory optimization

Languages: French (native), English (TOEFL 109), Mooré (fluent), German (basic)

PUBLICATIONS

Xu, W., Poussi, E., et al. *PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery*. Preprint, 2026. DOI: [10.64898/2026.02.26.707870](https://doi.org/10.64898/2026.02.26.707870)

EXTRACURRICULAR ACTIVITIES

Volunteering: Stanford BGSA Fundraising Chair (2025–Present) · Cheer Up cancer nonprofit (2023–2025) · XProjets consulting PM (2024–2025) · Math/physics tutor (2022–2023)

Sports: Handball (France's intercollegiate championship), Track & Field, Weightlifting **Arts:** Drawing, Painting, Dancing

Awards: French Government Excellence Scholarship (Ecole Polytechnique) · 2nd Place, 2021 Youth Olympiads, French Legion of Honor · 1st nationwide, 2020 Burkina Faso Scientific A-levels · 2020 Model Student Award · 2020–2018 Excellence Awards